Decoding Engagement in Live Lectures with EEG and Eye-Tracking

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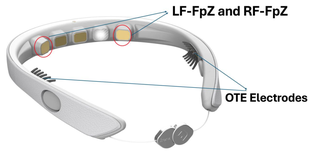
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INTRODUCTION

- Traditional measures (self-report, expert evaluation) do not directly capture students' engagement. Advances in EEG and eye-tracking address this limitation.
- However, findings from widely used video-based and virtual learning environments provide limited evidence on the relationship between engagement and auditory attention, restricting generalizability to classrooms.
- EEG responses to live speech remain incompletely understood.
- Classroom settings offer a critical context to study live speech processing and engagement decoding.

METHODS







Multimodal Data Collection (N = 10)

- Wearable EEG headband: FRENZ™ (Earable™ Neuroscience)
- Eye-tracking: Neon eye trackers
- Scalp EEG: 9 fronto-central channels: F3/4,Fz, FC3/4, FCz, C3/4, Cz

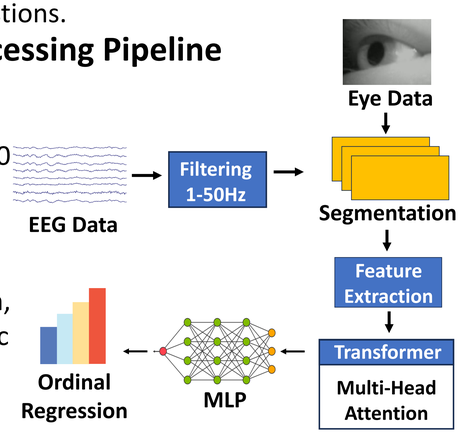
Procedure

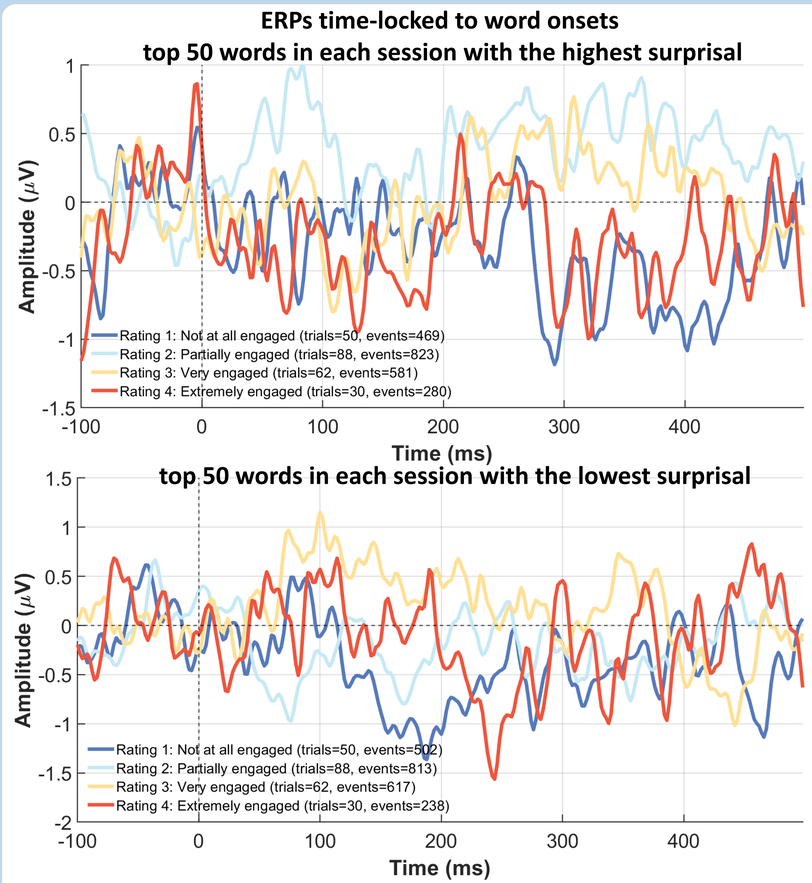
Four live lectures (12–18 minutes each)

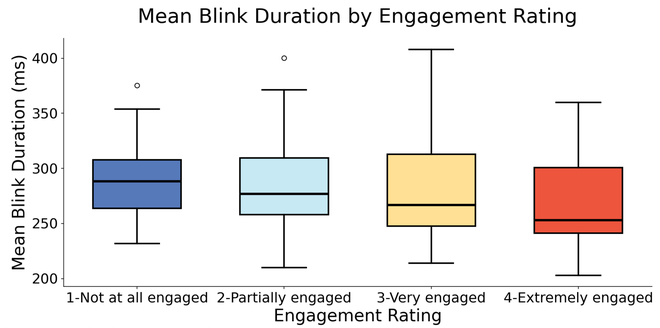
- First two lectures (rated as more interesting by the participant): delivered as live speech in an engaging style.
- Last two lectures (rated as less interesting by the participant): delivered as live speech in a boring style.
- During each lecture, participants were prompted to rate their engagement level five times. After each lecture, participants completed multiple-choice questions.

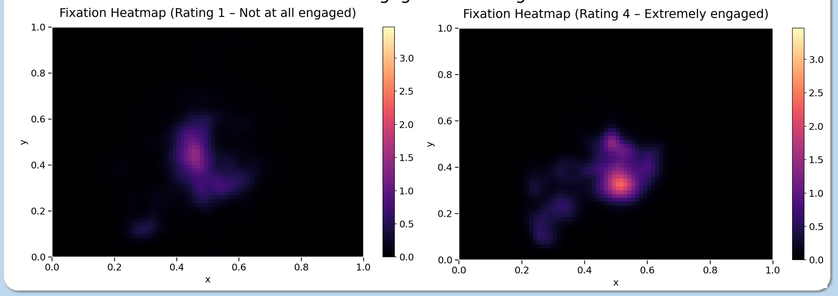
ERP Analysis and Data Processing Pipeline

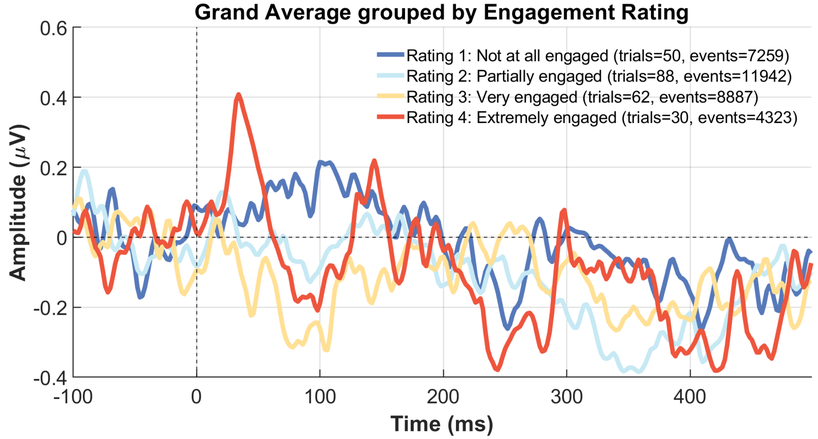
- Extract the top 750 acoustic onset events with the largest onset peak values and the top 50 words with the highest contextual word suprisal from each lecture recording.
- For each engagement rating bin, time-lock EEG epochs to acoustic onset peaks and word onsets and compute averaged ERPs.











Modality	Feature	MAE ↓	1-off ↑
Eye-tracking	Raw	0.888	0.850
	Blink	**0.708**	0.867
	Fixation	0.863	0.850
	Gaze	0.779	**0.875**
	Saccade	0.867	0.825
	Fix+Blink+Gaze	0.742	0.875
	Fix+Blink+Gaze+Saccade	1.000	0.729
Scalp-EEG	Bandpower	0.888	0.808
	Engagement Index	0.804	0.850
	Mean + SD	0.825	0.825
	ERP Amp	**0.789**	**0.871**
	All features	0.996	0.756
Headband	Bandpower	0.895	0.848
	Engagement Index	0.826	0.826
	Mean + SD	0.847	0.820
	All features	**0.802**	**0.865**

Conclusion

- The highly engaged condition elicited the strongest P1–N1–P2 ERP components to live speech.
- Cross-modality comparison between scalp and mobile EEG supports the feasibility of decoding engagement in real-world classroom settings.
- Mobile headband EEG demonstrates comparable engagement decoding performance to scalp EEG, further confirming its feasibility.

If you have any questions or would like to discuss this research further, feel free to reach out to me at **yimengw@umass.edu**. I'd be happy to chat and explore ideas together!